MILITARY SPECIFICATION

BROACH, PULP CANAL, BARBED, STERILE

This specification is approved for use within DLA, Defense Personnel Support Center (DM), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile barbed pulp canal broaches suitable for use in endodontic procedures.

1.2 Classification. The barbed pulp canal broaches shall be furnished in the following sizes, as specified (see 6.1):

Size XF - Extra Fine
Size M - Medium
Size C - Coarse

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

PPP-B-566	- Boxes, Folding Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Military

MIL-P-116	- Preservation, Method of.
MIL-S-36586	- Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

Federal

FED-STD-151	- Metal; Test Methods.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-government document which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 899 - Standard Specification for Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Division of Public Information, Food and Drug Administration, Washington, DC 20201).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Operative end. The operative end of the broaches shall be fabricated of ASTM type 301 or 302 austenitic stainless steel with a minimum tensile strength of 249,000 pounds per square inch (p.s.i.), when tested as specified in 4.4.1.

3.1.2 Handle. The handle of the broaches shall be fabricated of plastic material which is rigid and of a quality suitable to withstand normal operative procedures. Handles shall be capable of being autoclaved or dry heat sterilized without losing the shape of the handle or leaching out of color, when tested as specified in 4.4.8.

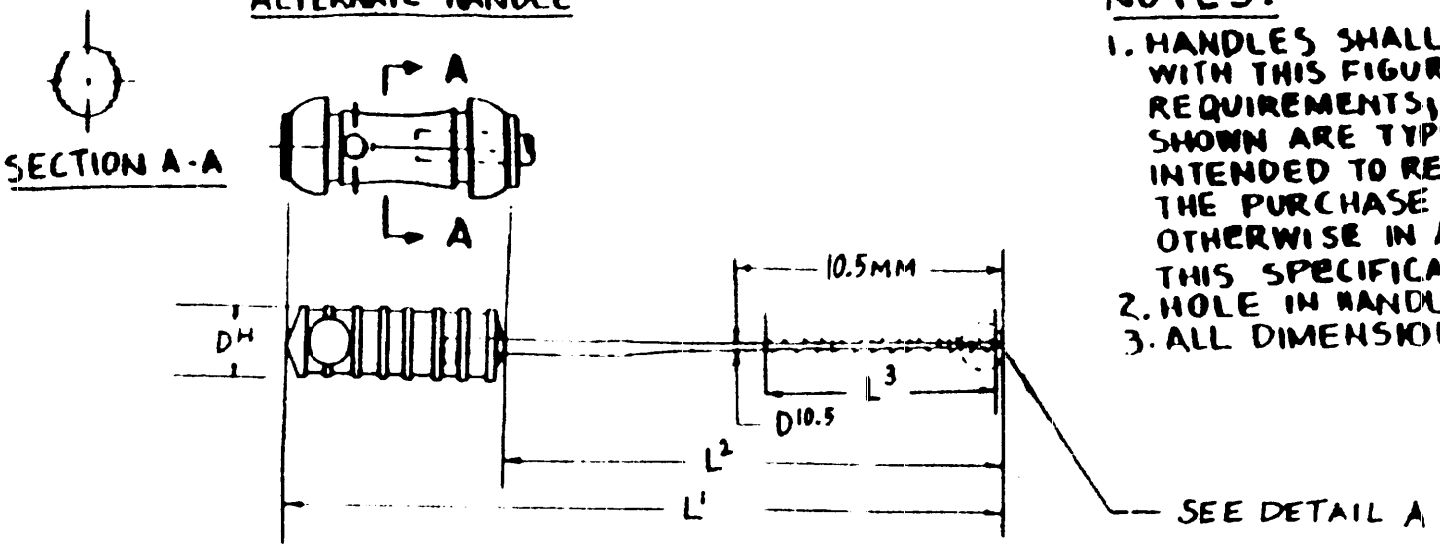
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3.2 Design and dimensions. Barbed broaches shall be of two-piece construction, consisting of an operative end (working end, neck, and shank) and a handle. Operative end shall be securely and permanently affixed into the handle with no movement of the operative end permitted. The barbed broaches shall be suitable for use in extirpating pulpal tissue and debridement of pulp canals during endodontic procedures. The broaches shall be straight. The full diameter of the wire shall be present over the entire length of the shank portion of the broach. The working end shall include a barbed section containing a minimum of 36 barbs when tested as specified in 4.4.2.2 The barbs shall be properly cut to a depth of 10 to 15 percent (%) of the diameter of the working end. The cross section of each barb shall increase parabolically from the tip to the base of the barb. Broaches shall be tapered for their full working length, with barbs arranged in uniformly spaced helical rows, projecting from the working end. Broaches shall be furnished in three sizes (see 6.1). Size XF shall be extra-fine; size M shall be medium; and size C shall be coarse. Design and dimensions of the broaches shall be as shown in Figure 1 and Table I.

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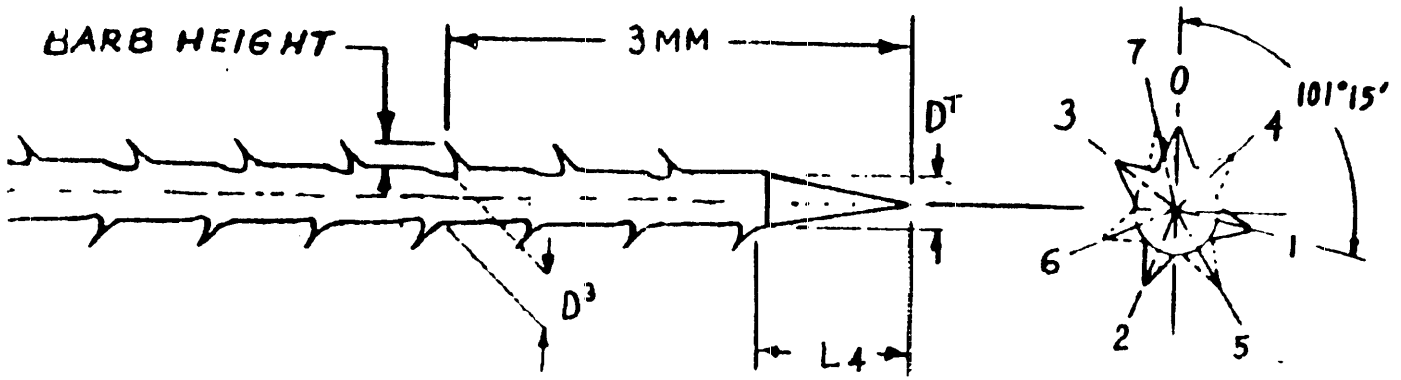
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ALTERNATE HANDLE



NOTES:

1. HANDLES SHALL BE IN ACCORDANCE WITH THIS FIGURE FOR DIMENSIONAL REQUIREMENTS; SHAPE OF HANDLES SHOWN ARE TYPICAL AND ARE NOT INTENDED TO RESTRICT OR PRECLUDE THE PURCHASE OF BROACHES WHICH ARE OTHERWISE IN ACCORDANCE WITH THIS SPECIFICATION.
2. HOLE IN HANDLE SHALL BE OPTIONAL.
3. ALL DIMENSIONS ARE IN MILLIMETERS.



DETAIL A (ENLARGED)

(TIP LENGTH)

NOTE:

BARB DISTANCE IS 101°15' WITH A STEADY SPIRAL INCLINATION OF 0.9 MM REF. BY 360 IN SEQUENCE 1 THRU 7 AS SHOWN.

FIGURE 1

TABLE I. Dimensions *

Size	$D^T \pm 0.03$	$D^3 \pm 0.03$	$D^{10.5} \pm 0.03$	$D^H \pm 0.10$	$L^1 \pm 1$	$L^2 \pm 1$	$L^3 \pm 0.2$	L^4 (max)	Barb Height ± 0.04
XF	0.18	0.21	0.28	3.0	32	22.5	9.0	1.0	0.11
M	0.25	0.28	0.35	3.0	32	22.5	9.0	1.0	0.14
C	0.29	0.32	0.39	3.0	32	22.5	9.0	1.0	0.16

*All dimensions are in millimeters.

D^T refers to the tip core diameter at 1 mm from tip.
D^3 refers to the core diameter at 3 mm from tip.
$D^{10.5}$ refers to the core diameter at 10.5 mm from tip.
D^H refers to the handle diameter.
L^1 refers to the total length of broach.
L^2 refers to the length of the operative end.
L^3 refers to the length of the barbed section.
L^4 refers to the length from the tip of the broach to the base of the first barb.

NOTE: Barb height is based on formula $D^3/2$ for each side broach.

3.2.1 Figures. The figures forming a part of this specification are specification control figures. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the figure to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2.2 Taper. The taper of the operative end shall be uniform from D3 to a position not less than 14 mm from the tip when tested as specified in 4.4.2.2. The taper of the barbed broach shall be .007, + .003, -0 mm per mm of length.

3.2.3 Tip. The tip length of the broach shall be no greater than the values given in Table I when tested as specified in 4.4.2.3. The shape of the tip shall be optional.

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3.2.4 Handles. Operative ends of the broaches shall be securely and permanently affixed in the handle when tested as specified in 4.4.3. Color shall be incorporated in the plastic composition. Handles shall be in accordance with figure 1 for dimensions.

3.3 Resistance to fracture by twisting. The maximum torsional load before failure and the angular deflection at failure for each broach shall not be less than that given in Table II for each size when tested as specified in 4.4.4.

3.4 Stiffness. The stiffness of all broaches shall not be greater than that given in Table III for each size when tested as specified in 4.4.5.

Table II.

Twist Torque and Angular Deflection (Minimum Values) Twisted in Clockwise Rotation.

Size	Torque (gm. cm.)	Angular Deflection (Degrees)
XF	10	90
M	20	90
C	35	90

Table III.

Stiffness (Bending Moment) Maximum Values at 45 Bend.

Size	Bending Torque (gm. cm.)
XF	50
M	120
C	170

3.5 Sterilization. The barbed broaches shall be gamma-ray sterilized. Sterility shall be tested as specified in 4.4.6.

3.6 Corrosion resistance. The operative ends of the broaches shall not show evidence of corrosion on their surface when subjected to the corrosion tests as specified in 4.4.7.

3.7 Finish. Barbed section of operative end of broaches shall have a bright and smoothly cut finish. Handles shall be longitudinally serrated or shall have cross cut serrations. The entire handle may be serrated or only the finger-gripping portion need be serrated. Handles shall be suitable for proper finger-gripping.

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3.8 Identification marking. Handles shall be color coded according to the applicable size as follows:

Size	Color
XF	Red
M	Green
C	Black

The color shall be impregnated into the plastic formulation. The presence of a letter or word on the handle corresponding to the size of the instrument shall be optional.

3.9 Workmanship. The entire broach shall be clean and free of foreign material, laminations, cracks, gouges, chips, burrs, broken areas and other defects which may detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables IV, V, and VI (see 4.2.1, 4.2.2 and 4.3) and 4.5.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table IV. Unit of product for examination purposes shall be the unit of issue.

TABLE IV. Sampling for examination.

Type Defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table V. Unit of product for test purposes shall be as indicated in table V.

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TABLE V. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Unit of product	Inspection level	AQL (percent defective)
Operative end	Tensile strength	3.1.1	4.4.1	Lot of steel stock	*	---
Entire Instrument	Dimensions	3.2, 3.2.3	4.4.2, 4.4.2.4	One broach	S-2	1.0
Entire Instrument	Handle security	3.2.4	4.4.3	One broach	S-2	1.0
Operative end	Resist. to frac. by twisting	3.3	4.4.4	One broach	S-2	1.0
Operative end	Stiffness test	3.4	4.4.5	One broach	S-2	1.0
Entire Instrument	Sterility test	3.5	4.4.6	One broach	S-2	**
Entire Instrument	Corrosion Resistance	3.6	4.4.7	One broach	S-2	1.0
Handles	Sterilization	3.1.2	4.4.8	One broach	***	2.5

*Average of a mininum of three specimens from each lot of steel stock.
**Acceptance number shall be zero for all sample sizes.
***Separate S-2 level quantities for both autoclave and dry heat tests.

4.2.3 Raw materials. Sufficient material from each lot of metal used in the construction of the broaches shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be tested for the determination of two or more test characteristics.

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4.3 Examination. The broaches shall be examined for defects including, but not limited to, those listed in table VI.

TABLE VI. Classification of Defects.

Categories	Defects
Major:	
101	Working end and neck of broach not tapered as specified.
102	Broach not free from cracks, grind marks, seams, pits, laminations or other defects which may impair serviceability.
103	Broach not barbed.
104	Style and design not as specified.
105	Barbs not arranged in uniformly spaced helical rows.
106	Finish of operative end not as specified.
107	Operative end bent, deformed, or otherwise damaged.
108	Operative end not firmly affixed in handle.
109	Handle not properly color coded.
Minor:	
201	Rust or other corrosion present.
202	Finish of handles not as specified.

4.4 Tests.

4.4.1 Chemical analysis and tensile strength of the metals. In the absence of certificates of quality for the raw material, the samples selected in accordance with 4.2.3 shall be analyzed in conformance with methods 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, method 111 shall be used for referee analysis. Tensile strength shall be performed in a tension tester on an 8-inch sample of steel stock used to produce the broaches.

4.4.2 Dimensions. The diameter taper, and tip length of a broach shall be measured using an optical comparator, shadowgraph, or other suitable device capable of measuring the diameter and length with the accuracy specified in 3.2 and Table I.

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4.4.2.1 Diameters. The broach to be measured shall be inserted into a suitable device and the diameters DT, D3, D10.5, and DH shall be measured.

4.4.2.2 Barbs. The number of barbs shall be determined when the branch is viewed under suitable magnification.

4.4.2.3 Taper. A broach to be measured shall have its diameter determined at 1 mm increments along the cutting section beginning 3 mm from the tip point up to 9 mm from the tip point.

4.4.2.4 Tip. The tip length of the instrument shall be no greater than the distance to the first barbed when determined by a suitable measuring device.

4.4.3 Handle security. The tested broaches shall have no axial movement from the handle when pulled with a minimum force of 2.25 kg. See figure 2 for suggested test equipment. The tested broaches shall not twist within the handle when torqued 360 gm cm maximum.

4.4.4 Resistance to fracture by twisting. Samples of a given size shall be tested for resistance to fracture by twisting as follows:

4.4.4.1 Equipment. The testing equipment shall be capable of measuring torque with an accuracy of $\pm$ 1 gm. cm. and angular deflection with an accuracy of $\pm$ 2°. The equipment described below shall be used for compliance testing and is presented as the preferred type of equipment for all purposes. The equipment as shown in Figure 3 shall consist of: a reversible geared motor revolving at a speed of 2 RPM which drives a test instrument by means of a separate hardened steel chuck and torque measuring device fixed on two linear ball bearings. Mounted to the shaft of the device is the chuck with jaws made of brass (70% copper-35% zinc) used to clamp the test instrument at the tip along a length of 3 mm. See Figure 4. A separate amplifier and digital display controls the operation of the motor and records the torque and angular deflection. The circuitry of the amplifier should include the feature of retaining for display the maximum torque 1 mm from the tip point up to 6 mm from the tip point.

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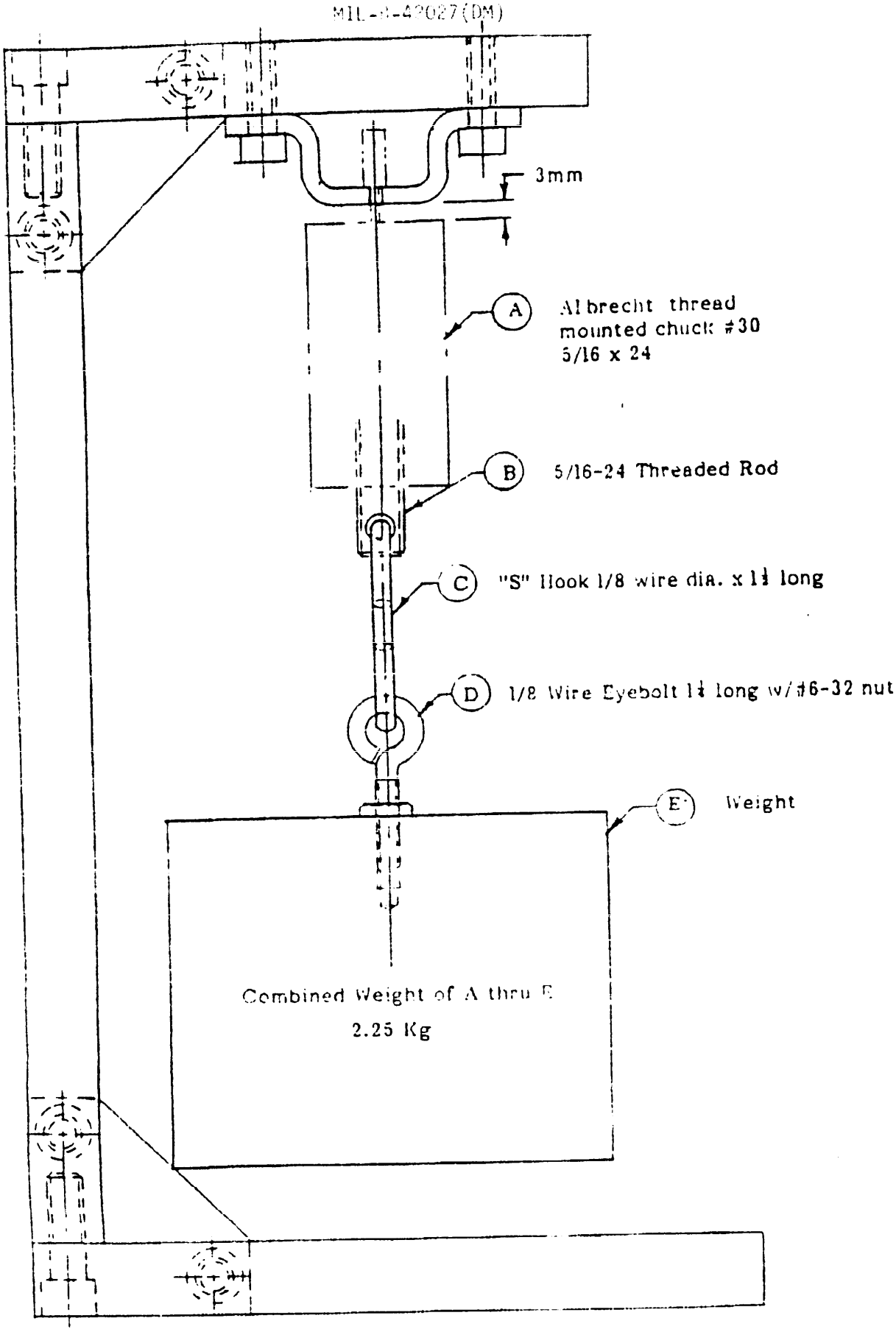


FIG. 2 - TEST EQUIPMENT FOR HANDLE SECURITY TESTING

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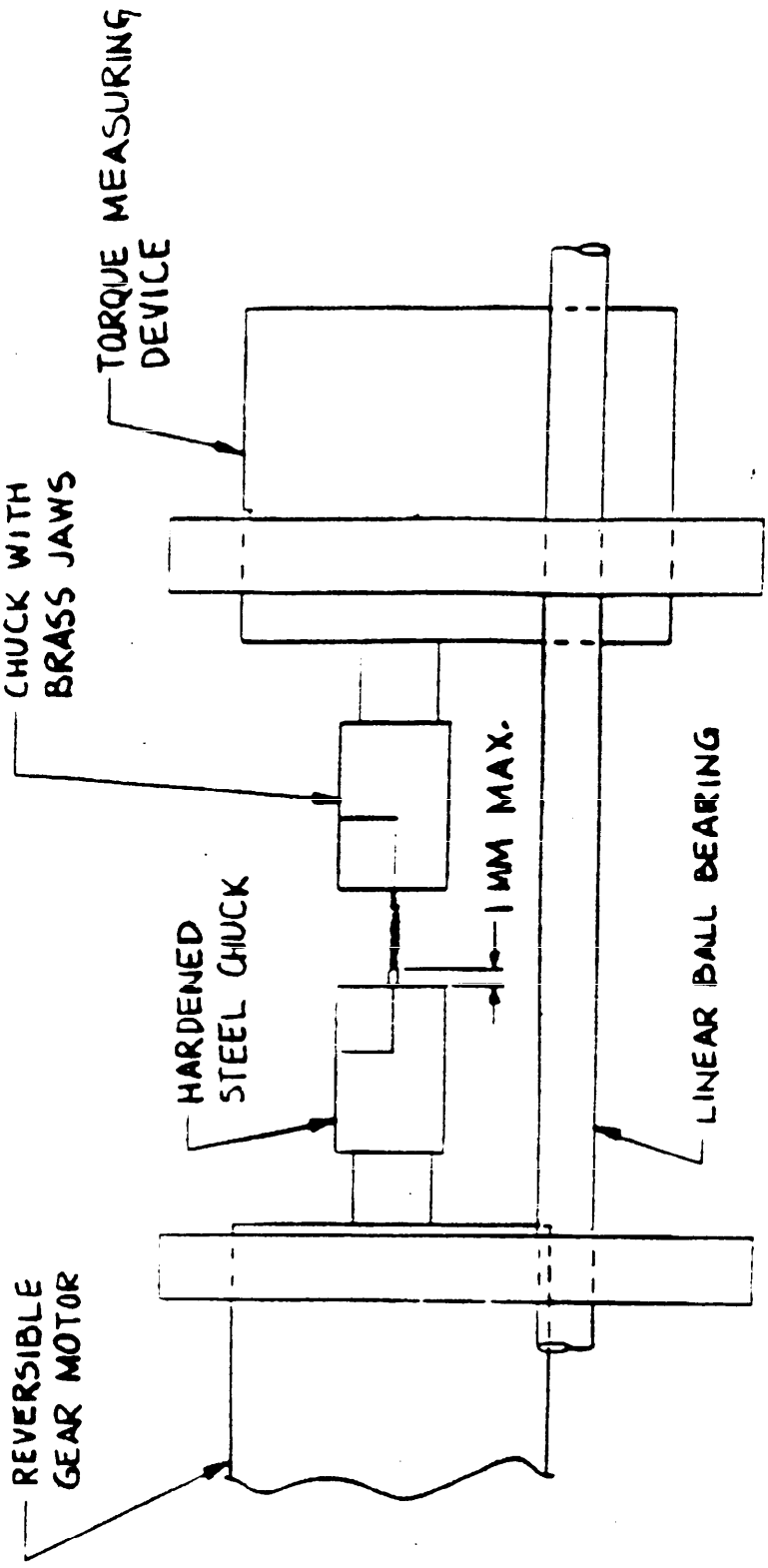


FIG. 3 - TEST EQUIPMENT FOR TORQUE TESTING

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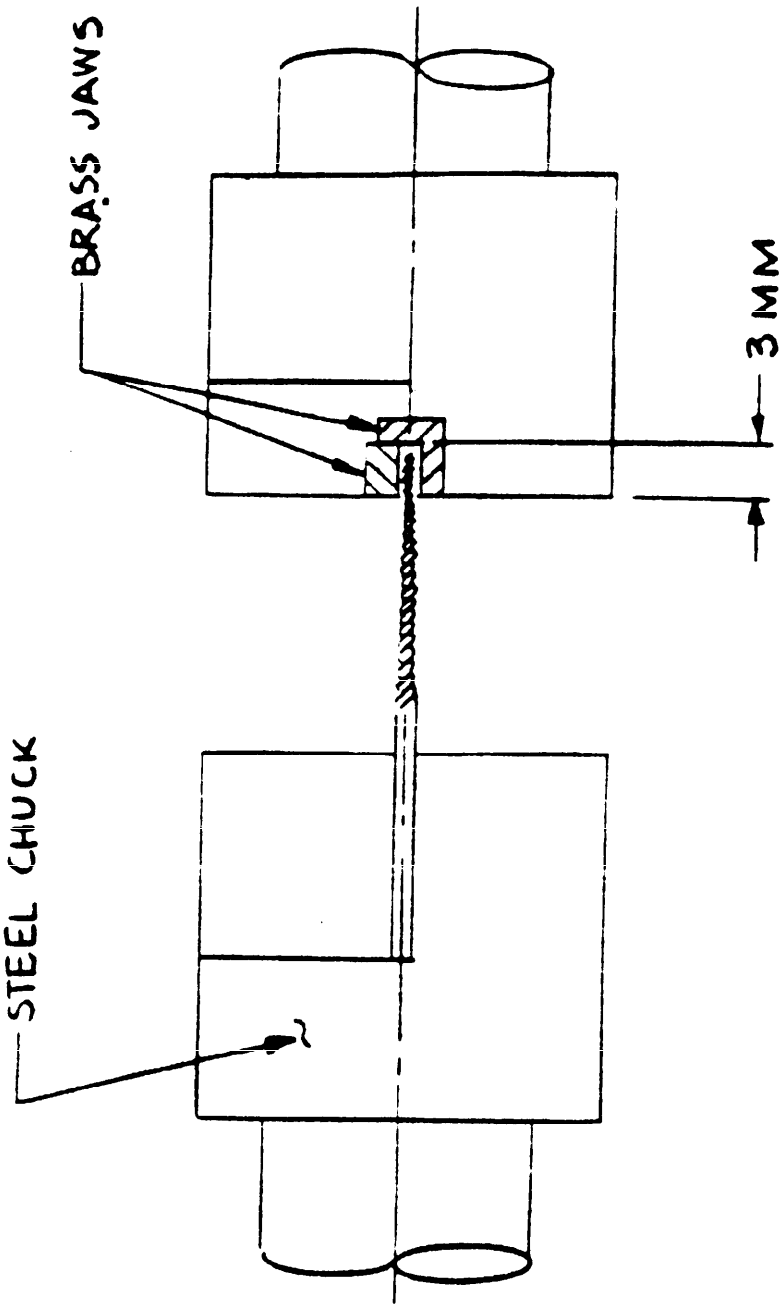


FIG. 4 - TWIST TEST CHUCK DETAIL

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4.4.4.2 Procedure for twist test. Calibrate the device for the torque range of the test instrument size to be tested. The broach to be tested shall have its handle removed with a suitable wire cutter at the point at which the handle is attached to the instrument shaft. Set the test instrument shank into the driving chuck leaving a maximum of 1 mm of the unground portion of the shank out of the chuck. Tighten the chuck. Slowly slide the torque device along the linear ball bearing until the tip of the test instrument enters 3 mm into the brass jaws. Check to assure that the test instrument is straight and centered into the jaws. Tighten the chuck. Since clamping will probably induce a pre-stress of the test instrument, the gear motor must be activated in steps until the torque digital display or the strip chart recorder shows a zero reading. After insuring that the geared motor is set for clockwise rotation as viewed from the test instrument shank end, activate the device. The test device will cease operation when the test instrument fails. The maximum torsional load before failure and the angular deflection at failure shall be recorded for each broach.

4.4.5 Stiffness. Samples of a given size shall be tested for stiffness as follows:

4.4.5.1 Equipment. The equipment as described in 4.4.4.1 shall be used with the modification of the clamping jaws and the bending device or catch pin as shown in Figure 5. The amplifier of the device must be capable of obtaining a pre-selected angular deflection (45°) at which point the test stops. The testing equipment shall be capable of measuring torque with an accuracy of ± 1 gm. cm.

4.4.5.2 Procedure for stiffness test. Calibrate the device for the torque range of the test instrument size to be tested. The broaches to be tested shall have its handle removed with a suitable wire cutter at the point where the handle is attached to the instrument shaft. Set the device to stop the angular deflection at 45°. Place the chuck onto the shaft of the torque meter device. Set the tip of the test instrument into the jaws of the chuck perpendicular to the axis of the motor. When the tip enters 3 mm into the brass jaws, tighten the chuck. Mount the catch pin into the gear motor shaft. Slide the torque device along the linear ball bearing until the test instrument is located above the rotating pin. By rotating the gear motor in the proper direction in steps, adjust the catch pin until it is lightly touching the test instrument. The torque reading must show zero. Activate the test device. The test will automatically cease when the angular deflection has reached 45°. The torque at 45° shall be recorded for each broach tested.

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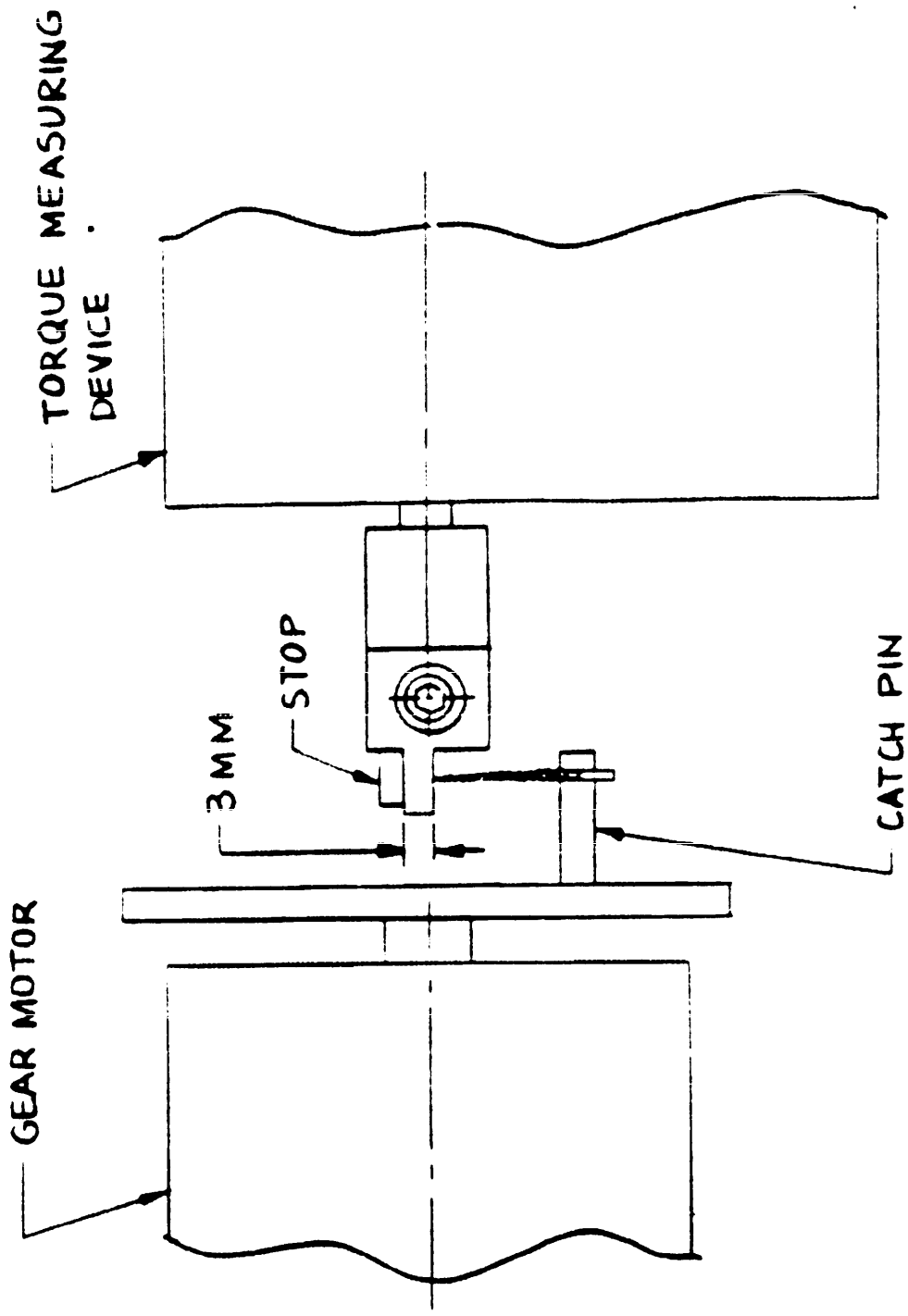


FIG. 5 - STIFFNESS TEST

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4.4.6 Sterility test. The broaches shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table V.

4.4.6.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. Stearothermophilus	55° - 65°C
Dry Heat	B. Stearothermophilus	55° - 65°C
	B. Subtilis var. niger (or globigii)	30° - 35°C
Gas	B. Subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. Pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.6.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6. Sterility Test and 4.4.6.1. Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP and paragraphs 4.4.6 and 4.4.6.1.

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4.4.6.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.6.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the USP. The inspection level shall be as indicated in table V.

4.4.7 Corrosion tests. Broaches shall be tested for corrosion resistance by one of the following procedures:

4.4.7.1 Sample preparation. Broaches of each size shall be scrubbed with soap and water, rinsed in hot water, followed by a dip in ethyl alcohol (95%), or isopropyl alcohol (isopropanol; 2-propanol) NF, and dried.

4.4.7.2 Copper sulfate test. A copper sulfate solution is made up as follows: Copper sulfate ($CuSO_4$, 5 H_2O) 4.0 g, Sulfuric acid (H_2SO_4, sp gr 1.84) 10.0 g, and Water (H_2O) 90.0 milliliters (ml). The instruments selected and prepared shall be immersed in the copper sulfate solution at a temperature of 65° Fahrenheit (F) for 6 minutes and then wiped off with a cloth saturated with fresh water. The instrument shall then be examined under a magnification of at least 5X for the presence of brown stains indicating corrosion.

4.4.7.3 Agar ion test. An agar ion gel is made up as follows: Potassium ferricyanide ($K_3Fe(CN)_6$) 0.2 g, Sodium chloride (NaCl) 1.0 g, Water (H_2O) 100 ml, Agar-agar 1.5 g. The agar sol shall be poured into a transparent container. The instruments selected and prepared are inserted into the gel (37° ± 2° Celsius (C)) at intervals of not less than 10 mm and at a distance of 10 mm from the periphery of the container. The instruments are inserted to a depth of aproximately 20 mm. Visual examination of the instruments while in the gel shall be made for the presence of blue precipitates on the instrument, indicating corrosion.

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4.4.8 Sterilization tests. The finished broaches shall be autoclaved under 27 ± 3 pounds of pressure at 132 ± 5°C (270 ± 9°F) for twenty (20) minutes and dry heat sterilized at 160 ± 5°C (320 ± 9° F) for one hundred and twenty (120) minutes. The handles of the broaches shall then be examined for compliance with specification requirements.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each broach shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Sterile broaches shall be packaged on vacuum formed plastic cards of sufficient clarity to identify the color and shape of the handle. Each card shall contain 10 cells. Each cell shall be protected by foil which allows easy removal of one instrument at a time and is capable of maintaining sterility of contents unless damaged or opened. Thirty broaches (3 cards of 10 broaches each) of one size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.1.4 Intermediate package. Ten unit packages containing broaches of one size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for broaches shall conform to the requirements of MIL-STD-794 for level C.

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5.1.2.2 Intermediate package. Ten unit packages containing broaches of one size only shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Seventy-two intermediate packages (720 units) containing broaches of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Seventy-two intermediate packages (720 units) containing broaches of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. Seventy-two intermediate packages (720 units) containing broaches of one size only shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Broaches packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Levels C. Broaches packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

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5.3 Marking.

5.3.1 Plastic card of 10 broaches. Each plastic card of 10 broaches shall be marked in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Marking shall include the lot (control) number and the legend "Sterility guaranteed unless package is damaged or opened" or similar commercial legend.

5.3.2 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129. Marking shall include the lot (control) number.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog.

National Stock Number	Item Identification
	BROACH, PULP CANAL, BARBED
6520-01-221-9089	Medium, Straight, 30: Sterile
6520-01-221-9090	Coarse, Straight, 30: Sterile
6520-01-222-9974	Extra-Fine, Straight, 30: Sterile

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6.3 Subject term (key word) listing.
Barbed
Broach
Pulp Canal
Sterile

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

Review Activities:

Army-MD
Navy-MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

60040
60041
60042

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6529-1975

Location: NCR DISK # 7

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-42027(DM)	BROACH, PULP CANAL, BARBED, STERILE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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